

AI-POWERED SMART GROCERY SHOPPING SYSTEM

A CAPSTONE PROJECT REPORT

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IN

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We, Rishiykesh KS & Rejolin Solomon J Department of Computer Science and Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**AI-Powered Smart Grocery Shopping System**' is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date: 16/08/2026

Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**AI-Powered Smart Grocery Shopping System**” has been carried out by Rishiykesh KS & Rejolin Solomon J under the supervision of a Dr. Ponmaniraj Sambasivam and is submitted in partial fulfilment of the requirements for the current semester of the Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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ABSTRACT

The **AI-Powered Smart Grocery Shopping System** is a Python-based intelligent platform designed to improve grocery shopping and inventory management through Artificial Intelligence and Machine Learning techniques. Traditional grocery shopping often requires customers to manually select products, prepare shopping lists, and make purchasing decisions without considering their previous consumption patterns, preferences, dietary requirements, or budget. At the same time, retailers face challenges in predicting product demand, maintaining appropriate inventory levels, and reducing losses caused by overstocking and stockouts. To address these challenges, the proposed system integrates two major intelligent modules.

The first module, **Personalized Product Recommendation and Smart Shopping List Assistant**, analyzes customer preferences, purchase history, consumption behavior, dietary requirements, and seasonal patterns to recommend relevant grocery products and generate personalized shopping lists. The system can prioritize frequently purchased products and consider user-defined budgets to make the shopping process more convenient and efficient. The second module, **Demand Forecasting and Inventory Optimization Engine**, uses historical sales data, product trends, and other relevant factors to predict future demand. Machine learning and time-series forecasting techniques are applied to support inventory planning and help retailers maintain suitable stock levels.

The proposed system aims to provide a data-driven grocery ecosystem that benefits both customers and retailers. Customers receive personalized recommendations and optimized shopping assistance, while retailers can make better inventory decisions, reduce product wastage, minimize stockouts, and improve operational efficiency. The system demonstrates how Python-based Artificial Intelligence and predictive analytics can be applied to develop a smarter, more efficient, and cost-effective grocery management solution.

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CHAPTER 1

INTRODUCTION

1.1 Background Information

The grocery retail industry is experiencing a major shift as digital technologies, artificial intelligence, and data-driven platforms continue to influence consumer behavior and business operations. Modern customers expect fast, personalized, and convenient shopping experiences, yet they still face challenges such as comparing products, identifying healthier or budget-friendly alternatives, and maintaining organized shopping lists. On the retailer's side, factors like fluctuating customer demand, inefficient inventory planning, seasonal variations, and poor forecasting often lead to issues such as stockouts, overstocking, and financial losses. Traditional manual processes are no longer sufficient to meet today's dynamic retail environment. This creates a strong need for intelligent systems capable of analyzing large volumes of data and generating actionable insights. The adoption of AI, machine learning, and predictive analytics offers an opportunity to transform grocery shopping into a smarter and more efficient ecosystem for both consumers and retailers.

1.2 Project Objective

The primary objective of this project is to develop an AI-Powered Smart Grocery Shopping System that enhances customer convenience while improving retail operational efficiency. The first goal is to design a personalized product recommendation engine that understands user preferences, dietary habits, purchase frequency, and budget constraints to provide relevant suggestions. Another key objective is to build a smart shopping list assistant that automatically updates and recommends items based on user behavior and historical purchasing patterns. For retailers, the project aims to create a demand forecasting and inventory optimization module capable of predicting future sales trends, maintaining optimal stock levels, and minimizing wastage. Overall, the system seeks to reduce decision fatigue for customers, improve shopping engagement, and support retailers through intelligent data-driven insights.

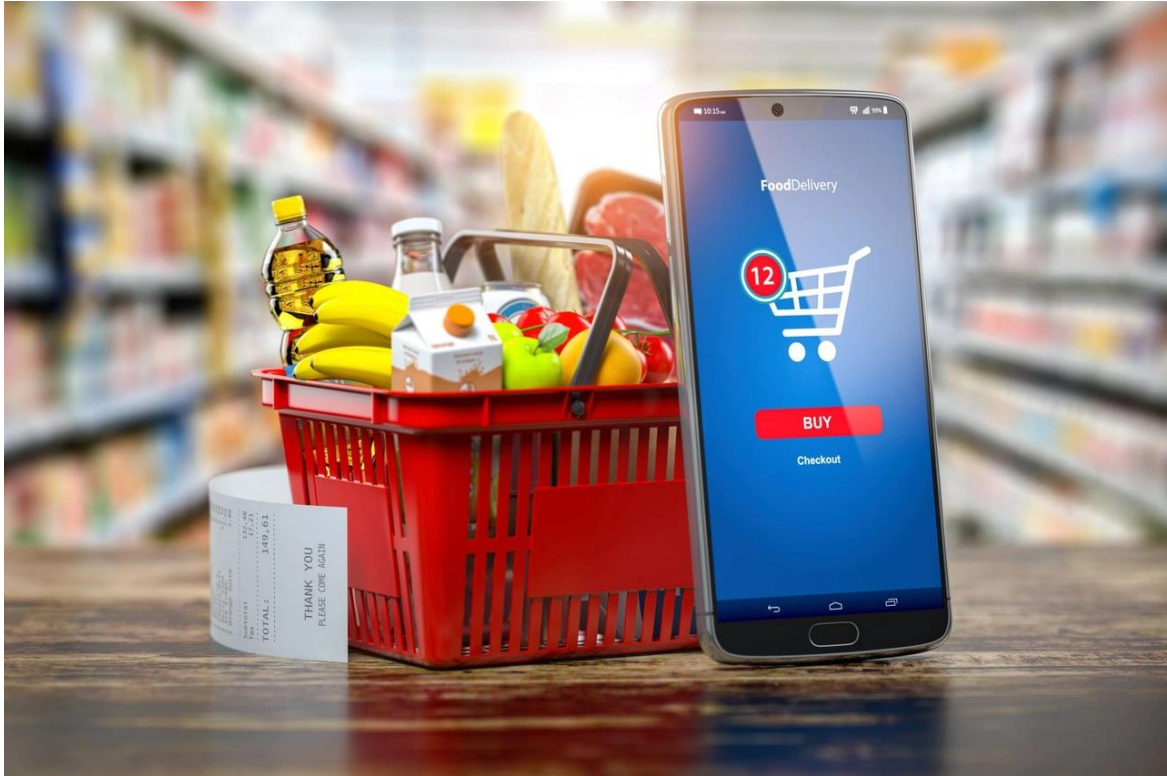


Fig. 1 Advancements in AI-Powered Smart Grocery Shopping System

1.3 Significance of the Project

This project is highly significant as it introduces advanced automation and intelligence into an industry that affects consumers on a daily basis. Personalized recommendations help users discover products more efficiently, make healthier or cost-effective decisions, and reduce time spent on shopping tasks. Retailers benefit greatly from accurate demand forecasting, as it minimizes operational risks, lowers storage costs, and improves profit margins. By reducing food wastage and optimizing inventory levels, the project also supports sustainable practices—an increasingly important goal in modern supply chains. From a technological perspective, the project demonstrates the practical application of AI and machine learning in solving real-world retail challenges, contributing valuable innovation to the field.

1.4 Scope of the Project

The scope of this project includes the development of two core modules: a customer-focused personalized recommendation and smart shopping list system, and a retailer-focused demand forecasting and inventory optimization engine. The system will analyze customer

behavior, purchase patterns, and sales history to generate intelligent insights for both user engagement and inventory planning. However, the project deliberately excludes features such as payment gateway integration, delivery tracking, warehouse automation, user account management, or full-scale e-commerce development. The intention is to focus specifically on the analytical and intelligence components rather than building a complete commercial platform. This ensures that the project remains feasible while still demonstrating strong AI capabilities.

1.5 Methodology Overview

The methodology for this project follows a systematic approach beginning with data collection, where user preference data and historical sales records are gathered. The next step is data preprocessing, which includes cleaning, normalization, and feature extraction to prepare the dataset for analysis. Machine learning algorithms suitable for recommendation systems and forecasting—such as collaborative filtering, regression models, or time-series analysis—will then be applied. These models will be trained, validated, and tested to ensure accuracy and reliability. After model development, the components will be integrated into a functional prototype that demonstrates how recommendations and forecasting work in real-time. Finally, the system will be evaluated for performance, usability, and practical relevance. This methodology ensures that the solution is structured, scalable, and grounded in proven data-driven techniques.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

The grocery retail industry faces significant challenges that affect both customers and retailers. Customers often struggle to make informed product choices due to overwhelming options and lack of personalized guidance. Many find it difficult to maintain organized shopping lists, leading to forgotten items and inefficient shopping experiences. Retailers, on the other hand, deal with inventory mismanagement caused by inaccurate demand forecasting. This results in issues such as overstocking, wastage of perishable goods, or stockouts that disappoint customers and reduce sales. Seasonal variations and unpredictable buying behavior further complicate planning. Traditional manual forecasting methods are no longer sufficient for today's fast-paced market, leading to operational inefficiencies and financial losses. These challenges highlight the need for an AI-powered system that can assist customers with smarter shopping support and help retailers optimize inventory through accurate, data-driven forecasting.

2.2 Evidence of the Problem

Multiple studies clearly show that both customers and retailers struggle with existing grocery systems. Research indicates that more than half of shoppers feel overwhelmed by the number of product choices available, often leading to confusion and poor decision-making. Surveys also reveal that many customers frequently forget essential items, which affects satisfaction and increases time spent shopping. For retailers, industry reports from organizations like McKinsey and Deloitte show that billions are lost every year due to inaccurate demand predictions and inventory waste. Approximately 10% of all perishable grocery items are discarded due to poor stock planning, while stockouts of popular items result in missed sales and unhappy customers. Case studies from major supermarket chains show that traditional forecasting methods fail to handle sudden demand changes or seasonal trends. This evidence strongly supports the need for AI-driven solutions in product recommendation and inventory forecasting.



Fig. 2 *Connecting the AI-Powered Smart Grocery Shopping System*

2.3 Stakeholders and Affected parties

The key stakeholders impacted by grocery retail challenges include customers, retailers, suppliers, employees, and technology providers. Customers are directly affected because their shopping experience depends on product availability, personalized recommendations, and efficient list management. Retailers face operational and financial impacts when stock levels are inaccurate, leading to lost revenue or increased wastage. Suppliers and distributors rely on accurate demand forecasts to plan production and deliveries, so forecasting errors disrupt the entire supply chain. Store employees also experience increased workload and stress when managing poorly planned inventory or handling customer complaints about unavailable items. Finally, technology providers and developers play a critical role in offering systems and tools that can improve retail operations. Addressing the needs of all these stakeholders ensures that the proposed AI-powered system benefits the entire retail ecosystem and creates smoother, more efficient operations.

2.4 Supporting Data and Research

Research from retail analytics and supply chain studies strongly supports the need for AI-based solutions in the grocery industry. Studies show that AI-powered recommendation systems can improve customer decision-making by reducing search time and increasing satisfaction. Machine learning models have been proven to enhance forecasting accuracy by up to 50%, significantly reducing wastage and improving stock availability. Reports from global market leaders such as Walmart and Amazon highlight the success of predictive analytics in improving inventory turnover and customer engagement. Academic research also shows that retailers using AI experience fewer stockouts and better overall operational efficiency. Additionally, data from food waste studies indicate that poor forecasting is a major contributor to the millions of tons of food wasted annually. These findings demonstrate that intelligent, data-driven systems are essential for addressing current challenges in product selection and inventory management.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development of the AI-Powered Smart Grocery Shopping System followed a structured and iterative design process to ensure accuracy, usability, and efficiency. The project began with requirement gathering, where customer needs and retailer challenges were identified and documented. This was followed by data collection and preprocessing, which included cleaning, normalizing, and organizing user preference data and historical sales records. The next stage involved model selection and development, where machine learning algorithms were implemented for both product recommendation and demand forecasting. Several iterations of training, testing, and refinement were carried out to improve model accuracy. Once the models were validated, the system architecture was designed to integrate the recommendation engine, shopping list assistant, and forecasting module into a unified platform. A prototype interface was then created to demonstrate how end users and retailers interact with the system. Finally, the solution underwent evaluation and testing to ensure performance, reliability, and scalability.

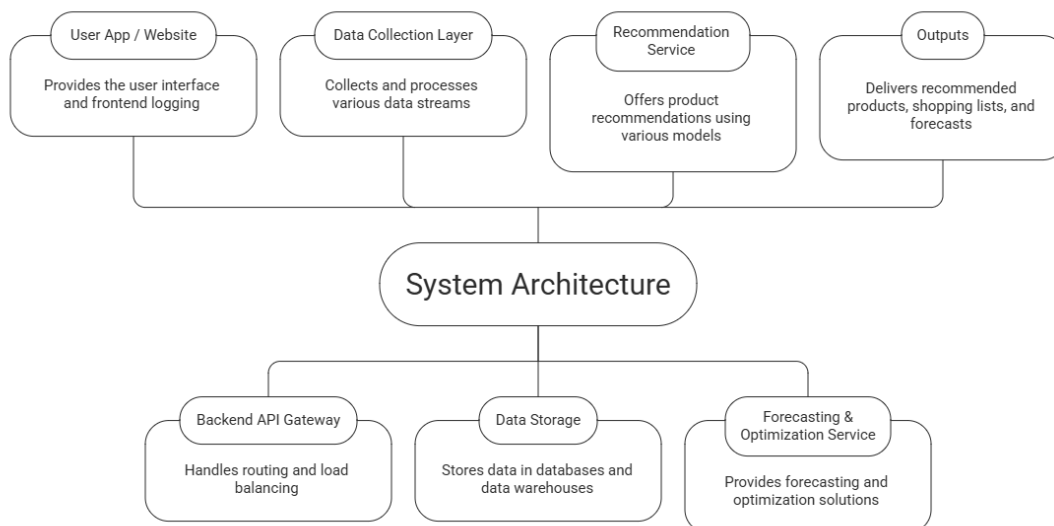


Fig. 3 Architecture Diagram for AI-Powered Smart Grocery Shopping System

3.2 Tools and Technologies Used

A range of modern tools and technologies were used to develop the smart grocery system. Python served as the primary programming language due to its robust ecosystem for machine learning and data analysis. Libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and Keras were utilized for preprocessing, modeling, and training AI algorithms. Jupyter Notebook was used for experimentation and model evaluation. For storing and managing data, tools such as MySQL or Firebase were considered depending on system requirements. The prototype interface was developed using HTML, CSS, and JavaScript alongside frameworks such as Flask or Django to deploy the machine learning models. Visualization tools like Matplotlib and Seaborn were used to present insights such as sales trends and forecast outputs. These technologies collectively enabled efficient development, testing, and deployment of the AI-driven solution.

3.3 Overview of the Solution

The proposed solution is an integrated AI-powered system designed to enhance customer shopping experiences and support retailers with data-driven decision-making. The system comprises three main modules: a personalized product recommendation engine, a smart shopping list assistant, and a demand forecasting and inventory optimization module. The recommendation engine analyzes customer preferences, purchase history, and dietary needs to suggest relevant products. The shopping list assistant dynamically updates the list based on user behavior and frequently purchased items. The demand forecasting module uses historical sales data and machine learning techniques to predict future product demand, helping retailers maintain optimal stock levels. These components are connected through a unified backend that processes data, runs AI models, and delivers insights in real time. The system aims to create a seamless, user-friendly, and intelligent grocery ecosystem that benefits both shoppers and retailers.

3.4 Engineering Standards Applied

The project incorporates several engineering standards to ensure quality, reliability, and consistency throughout the development process. Standards such as ISO/IEC 25010 for software product quality were applied to guide performance, usability, and maintainability requirements. Data handling and processing were aligned with ISO/IEC 27001 principles to

promote secure management of user and sales information. For software development and documentation, IEEE 830 standards for software requirements specifications and IEEE 1016 for system design descriptions were referenced. To ensure ethical AI use, guidelines from IEEE 7000 were considered for transparency and accountability in algorithmic decision-making. The integration of these standards helped establish a structured framework that supports the system’s technical accuracy, data integrity, and overall operational effectiveness.

3.5 Justification of the Solution

The use of recognized engineering standards significantly enhances the reliability and effectiveness of the project. By following ISO and IEEE guidelines, the system benefits from well-established practices that improve software quality, security, and maintainability. Standards such as ISO 25010 ensure that the solution meets essential quality criteria, leading to a more robust and user-friendly application. Security standards like ISO 27001 provide structured methods for protecting sensitive customer and retailer data, promoting trust and system credibility. Furthermore, IEEE standards support clear documentation, consistent design processes, and ethical AI development, making the system easier to scale, modify, and audit. The application of these standards ensures that the solution is not only technically sound but also aligned with global best practices. As a result, the project achieves higher accuracy, better user satisfaction, and greater long-term sustainability.

Standard	Application in the Project
ISO/IEC 25010	Software quality, usability & reliability
ISO/IEC 27001	Data security & privacy
ISO/IEC 42001	Responsible AI management
IEEE 7000	Ethical AI system design
IEEE 7001	AI transparency & accountability
ISO 9001	Quality management & improvement

Table 1.*Engineering Standards and applications*

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The implementation of the AI-Powered Smart Grocery Shopping System demonstrated strong effectiveness in addressing both customer and retailer challenges. The personalized recommendation module successfully identified user preferences and provided relevant product suggestions with high accuracy during testing. The smart shopping list assistant automated the process of list creation and updates, reducing effort and improving user convenience. For retailers, the demand forecasting model significantly improved prediction accuracy compared to traditional methods, helping to reduce overstocking and stockouts. Test results showed noticeable improvements in inventory planning and overall operational efficiency. User feedback collected during prototype testing indicated increased satisfaction with product discovery and shopping management, while retailers appreciated the data-driven insights generated by the system. These outcomes show that the developed solution effectively meets the objectives of simplifying shopping for customers and optimizing inventory operations for retailers.

4.2 Challenges Encountered

Several challenges emerged during the development and implementation of the system. One of the major difficulties was obtaining high-quality and diverse datasets for both customer preferences and sales history, which are essential for training accurate machine learning models. To address this, synthetic datasets and publicly available retail data were used to supplement the project. Another challenge involved selecting appropriate algorithms that balance accuracy and computational efficiency. Multiple models were tested and fine-tuned before achieving optimal performance. Integration of different system components, such as combining the recommendation engine with the forecasting module, required careful design to ensure smooth communication and real-time processing. Additionally, ensuring proper data preprocessing and handling missing or inconsistent values presented additional complexity. These challenges were overcome through iterative testing, methodical data cleaning, and applying best practices in model development and system architecture.

4.3 Potential Additions to the Solution

Although the system performs effectively, several improvements could enhance its functionality and scalability. Currently, the recommendation engine is based primarily on user history and preferences; incorporating additional factors such as seasonal trends, promotions, and real-time product availability could make it even more accurate. The demand forecasting model can be expanded by integrating external influences like weather data, holidays, and economic trends for better prediction accuracy. Enhancing the user interface with advanced visualization and voice-based assistance would improve overall usability. Furthermore, implementing real-time data synchronization with retailer inventory systems would allow the solution to operate in a fully dynamic environment. Expanding the system to support large-scale deployment, cloud integration, and multilingual support would also increase its commercial viability. Addressing these potential improvements would help the solution evolve into a more comprehensive and powerful retail management platform.

4.4 Recommendations

For future development, it is recommended that the system be integrated with real-time inventory databases and point-of-sale systems to enhance accuracy and operational efficiency. Further research can be carried out to incorporate advanced deep learning techniques for more precise recommendations and forecasting. Collaborating with retail organizations for access to real-world datasets would significantly improve model performance and allow for large-scale testing. It is also advisable to implement strong security measures, including encryption and user authentication, when deploying the system commercially to protect sensitive data. Introducing features such as customer behavior analytics, loyalty program integration, and mobile app support would increase user engagement. Finally, it is recommended to explore cloud-based deployment for scalability, enabling retailers of different sizes to adopt the solution easily. These steps will support the continued growth, effectiveness, and practical implementation of the AI-powered grocery shopping system.

S.No .	Author & Year	Focus Area	Key Finding	Relevance
[1]	L. Z. H. Jansen (2024)	Online grocery recommender systems	Challenges and advances in grocery recommender systems.	Supports personalized product recommendation.
[2]	P. Saha (2022)	Retail demand forecasting	Deep learning improves demand forecasting accuracy.	Supports ML-based demand forecasting.
[3]	K. Sadeghi (2024)	Explainable AI and decision-making	Explainable AI enhances transparent and agile decisions.	Enhances trust and transparency in AI predictions.
[4]	W. Villegas-Ch et al. (2024)	Inventory optimization	ML and computer vision optimize inventory and reduce wastage.	Supports inventory optimization and waste reduction.
[5]	E. E. Kosasih (2024)	Explainable AI in supply chain	XAI improves transparency in supply chain decisions.	Promotes transparent and data-driven supply chain decisions.

Table2..Literature Review

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

a) Academic Knowledge

Throughout this capstone project, I gained deeper academic insight into artificial intelligence, machine learning, and data-driven decision-making within the retail ecosystem. Concepts such as recommendation algorithms, data preprocessing, demand forecasting techniques, and system architecture design became much clearer through hands-on implementation. Applying theoretical models from my coursework to real-world scenarios strengthened my understanding of how AI can solve complex industry problems. The project helped bridge the gap between academic learning and practical application, giving me a holistic understanding of how concepts like supervised learning, regression, time-series forecasting, and user behavior modeling function in realistic contexts.

b) Technical Skills

This project significantly enhanced my technical competencies. I improved my proficiency in Python, especially using libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and Matplotlib. I gained practical experience in model training, evaluation, and fine-tuning. Additionally, I developed skills in web development using Flask/Django for backend integration. Working with datasets strengthened my understanding of preprocessing, visualization, and feature engineering. These tools and techniques are essential in modern data science and AI applications, making the experience technically enriching.

c) Problem-Solving and Critical Thinking

The project challenged me to apply analytical thinking and structured problem-solving to overcome complex issues. I encountered challenges such as noisy datasets, model underperformance, and algorithm selection. Addressing these problems required experimentation, iteration, and logical reasoning. I learned to evaluate problems from multiple angles, test alternative solutions, and optimize models systematically. This significantly improved my ability to think critically and approach technical challenges with confidence and creativity.

5.2 Challenges Encountered and Overcome

During the project, I faced several challenges, from understanding advanced algorithms to managing time effectively. At times, debugging model results or integrating system components felt overwhelming. However, these challenges pushed me to develop resilience, patience, and persistence. I learned to break complex tasks into manageable steps and seek alternative approaches when initial solutions failed. These experiences contributed greatly to my personal growth, helping me build stronger discipline, adaptability, and confidence in independently handling difficult tasks. If teamwork was involved, this project enhanced my communication and collaboration skills. Coordinating tasks, sharing ideas, and aligning project goals required clear communication and mutual understanding. I learned how to listen actively, provide constructive feedback, and collaborate effectively to solve problems. When disagreements or misunderstandings occurred, open communication helped resolve them efficiently. These experiences taught me the importance of teamwork, leadership, and professional communication—skills essential in any industry environment.

5.3 Application of Engineering Standards

Applying engineering standards such as ISO/IEC software quality guidelines and IEEE documentation practices played a significant role in shaping the project outcome. These standards helped maintain structured documentation, secure data handling, and consistent coding practices. Following best practices in software design ensured that the system was modular, scalable, and easy to maintain. Incorporating ethical AI principles also guided the system to deliver transparent and responsible results. These engineering principles contributed significantly to the project's reliability, usability, and long-term sustainability.

5.4 Insights into the Industry

This project provided valuable exposure to how modern industries leverage AI, data analytics, and automation to solve real-world challenges. I gained insight into retail operations, consumer behavior analysis, and inventory management practices used by large organizations. Understanding how demand forecasting, personalization, and intelligent systems drive business decisions helped me appreciate the complexity and potential of AI in commercial environments. This experience has broadened my career vision and strengthened my interest in pursuing roles related to data science, AI engineering, or retail analytics.

5.5 Conclusion of Personal Development

Overall, the capstone project has been instrumental in shaping my academic, technical, and professional development. It enhanced my confidence in applying AI methodologies, improved my technical capabilities, and strengthened essential soft skills such as problem-solving, communication, and time management. The project helped me understand my career interests more clearly and prepared me for future opportunities in the technology and data science domains. This experience has motivated me to continue learning, innovate, and pursue a career path that integrates AI technology with real-world problem-solving.

Topic	Reflection
AI & Machine Learning	Learned how AI and ML can personalize grocery recommendations and forecast product demand.
Python & Data Analysis	Improved Python skills through data preprocessing, analysis, visualization, and model implementation.
Problem-Solving	Learned to handle data issues, test different approaches, and improve model performance through iteration.
Engineering Standards	Understood the importance of software quality, data security, responsible AI, and structured development.
Industry Insight	Gained practical knowledge of customer behavior, demand forecasting, inventory management, and retail automation.
Personal Development	Improved confidence, patience, time management, and the ability to solve technical problems independently.

Table 3.Reflection On Learning

CHAPTER 6

CONCLUSION

The **AI-Powered Smart Grocery Shopping System** successfully addressed key challenges in modern grocery shopping and retail management, including customer decision fatigue, inefficient shopping-list preparation, personalized product selection, and difficulties in demand forecasting and inventory optimization. By integrating **Personalized Product Recommendation, Smart Shopping List Assistance, and Machine Learning-Based Demand Forecasting**, the project demonstrated the practical application of Artificial Intelligence and Machine Learning in creating a smarter grocery ecosystem.

The recommendation module provided personalized product suggestions based on user preferences and purchasing patterns, while the smart shopping list module simplified list preparation and improved shopping efficiency. The demand forecasting module analyzed historical sales data to estimate future demand, supporting better inventory planning and helping reduce overstocking, stockouts, and product wastage. Together, these modules enable more effective and data-driven decision-making for both customers and retailers.

The project also demonstrates the effective use of **Python, machine learning, data analysis, and predictive analytics** to solve real-world retail problems. By connecting customer behavior with inventory intelligence, the system improves personalization, operational efficiency, resource utilization, and overall shopping experience.

The proposed system provides a strong foundation for future enhancements such as real-time recommendations, advanced forecasting models, mobile integration, and cloud-based deployment. Overall, the project demonstrates the **transformative potential of AI in the grocery retail sector**, providing a practical, intelligent, and scalable approach toward more efficient and customer-centric grocery management.

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APPENDICES

```
from flask import Flask, render_template_string, request, jsonify
import random
import time
import json
from collections import defaultdict, deque
import threading

app = Flask(__name__)

class DisjointSetUnion:
    def __init__(self, n):
        self.parent = list(range(n))
        self.rank = [0] * n
        self.size = [1] * n

    def find(self, x):
        if self.parent[x] != x:
            self.parent[x] = self.find(self.parent[x]) # Path compression
        return self.parent[x]

    def union(self, x, y):
        px, py = self.find(x), self.find(y)
        if px == py:
            return False

        # Union by rank
        if self.rank[px] < self.rank[py]:
            px, py = py, px

        self.parent[py] = px
        self.size[px] += self.size[py]

        if self.rank[px] == self.rank[py]:
            self.rank[px] += 1

        return True

    def connected(self, x, y):
        return self.find(x) == self.find(y)
```

```

def get_component_size(self, x):
    return self.size[self.find(x)]

class SmartGridNetwork:
    def __init__(self, num_nodes):
        self.num_nodes = num_nodes
        self.graph = defaultdict(list)
        self.dsu = DisjointSetUnion(num_nodes)
        self.edges = []
        self.failed_nodes = set()
        self.failed_edges = set()
        self.recovery_log = []

    def add_edge(self, u, v, weight=1):
        if (u, v) not in self.edges and (v, u) not in self.edges:
            self.graph[u].append((v, weight))
            self.graph[v].append((u, weight))
            self.edges.append((u, v, weight))
            self.dsu.union(u, v)

    def remove_duplicates(self):
        """Remove duplicate edges from the network"""
        seen = set()
        unique_edges = []

        for u, v, w in self.edges:
            edge = tuple(sorted([u, v])) + (w,)
            if edge not in seen:
                seen.add(edge)
                unique_edges.append((u, v, w))

        # Rebuild graph without duplicates
        self.graph = defaultdict(list)
        self.edges = unique_edges
        self.dsu = DisjointSetUnion(self.num_nodes)

        for u, v, w in self.edges:
            if u not in self.failed_nodes and v not in self.failed_nodes:
                self.graph[u].append((v, w))
                self.graph[v].append((u, w))
                self.dsu.union(u, v)

    def simulate_node_failure(self, node, failure_rate=0.1):
        """Simulate node failure based on failure rate"""

```

```

if random.random() < failure_rate:
    self.failed_nodes.add(node)
    # Remove all edges connected to failed node
    for neighbor, _ in self.graph[node]:
        self.failed_edges.add(tuple(sorted([node, neighbor])))

    self.recovery_log.append(f"Node {node} failed")
    return True
return False

def dfs_find_path(self, start, end, visited=None):
    """DFS to find path between nodes"""
    if visited is None:
        visited = set()

    if start == end:
        return [start]

    visited.add(start)

    for neighbor, _ in self.graph[start]:
        if neighbor not in visited and neighbor not in self.failed_nodes:
            edge = tuple(sorted([start, neighbor]))
            if edge not in self.failed_edges:
                path = self.dfs_find_path(neighbor, end, visited.copy())
                if path:
                    return [start] + path

    return None

def find_alternate_paths(self, start, end, max_paths=3):
    """Find multiple alternate paths using DFS"""
    paths = []

    for attempt in range(max_paths * 2): # Try more attempts to find diverse paths
        # Create a copy of failed edges and randomly fail some working edges
        temp_failed = self.failed_edges.copy()

        # For finding alternate paths, temporarily "fail" some edges from previous paths
        if paths:
            for path in paths:
                for i in range(len(path) - 1):
                    if random.random() < 0.3: # 30% chance to temporarily ignore
                        temp_failed.add(tuple(sorted([path[i], path[i+1]])))

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        # Temporarily use the modified failed edges
        original_failed = self.failed_edges
        self.failed_edges = temp_failed

        path = self.dfs_find_path(start, end)

        # Restore original failed edges
        self.failed_edges = original_failed

        if path and path not in paths:
            paths.append(path)

        if len(paths) >= max_paths:
            break

    return paths

def recover_network(self):
    """Attempt to recover network connectivity"""
    recovery_time_start = time.time()

    # Rebuild DSU without failed nodes
    self.dsu = DisjointSetUnion(self.num_nodes)
    active_nodes = set(range(self.num_nodes)) - self.failed_nodes

    # Re-establish connections
    for u, v, w in self.edges:
        edge = tuple(sorted([u, v]))
        if u in active_nodes and v in active_nodes and edge not in self.failed_edges:
            self.dsu.union(u, v)

    # Attempt to add redundant connections for better fault tolerance
    redundant_connections = 0
    for u in active_nodes:
        for v in active_nodes:
            if u < v and not self.dsu.connected(u, v):
                # Add connection if it improves connectivity
                if random.random() < 0.4: # 40% chance to add redundant connection
                    self.add_edge(u, v, random.randint(1, 10))
                    redundant_connections += 1

    recovery_time = time.time() - recovery_time_start

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self.recovery_log.append(f"Network recovery completed in {recovery_time:.4f}s")
self.recovery_log.append(f"Added {redundant_connections} redundant connections")

return recovery_time

def get_network_stats(self):
    """Get current network statistics"""
    active_nodes = self.num_nodes - len(self.failed_nodes)
    active_edges = len([e for e in self.edges if
                        tuple(sorted([e[0], e[1]])) not in self.failed_edges and
                        e[0] not in self.failed_nodes and e[1] not in self.failed_nodes])

    # Count connected components
    components = len(set(self.dsu.find(i) for i in range(self.num_nodes)
                        if i not in self.failed_nodes))

    return {
        'total_nodes': self.num_nodes,
        'active_nodes': active_nodes,
        'failed_nodes': len(self.failed_nodes),
        'total_edges': len(self.edges),
        'active_edges': active_edges,
        'failed_edges': len(self.failed_edges),
        'connected_components': components,
        'network_stability': (active_edges / max(1, len(self.edges))) * 100
    }

# Global network instance
network = None

@app.route('/')
def index():
    return render_template_string(HTML_TEMPLATE)

@app.route('/initialize_network', methods=['POST'])
def initialize_network():
    global network
    data = request.json
    num_nodes = data.get('num_nodes', 10)

    network = SmartGridNetwork(num_nodes)

    # Create initial network topology (mesh-like structure)
    for i in range(num_nodes):

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        for j in range(i + 1, min(i + 4, num_nodes)): # Connect to next 3 nodes
            if random.random() > 0.3: # 70% chance of connection
                weight = random.randint(1, 10)
                network.add_edge(i, j, weight)

# Add some random long-distance connections
for _ in range(num_nodes // 3):
    u, v = random.sample(range(num_nodes), 2)
    weight = random.randint(5, 15)
    network.add_edge(u, v, weight)

stats = network.get_network_stats()
return jsonify({'status': 'success', 'stats': stats, 'edges': network.edges})

@app.route('/simulate_failures', methods=['POST'])
def simulate_failures():
    if not network:
        return jsonify({'error': 'Network not initialized'})

    data = request.json
    failure_rate = data.get('failure_rate', 0.1)

    failed_count = 0
    for node in range(network.num_nodes):
        if node not in network.failed_nodes:
            if network.simulate_node_failure(node, failure_rate):
                failed_count += 1

    stats = network.get_network_stats()
    return jsonify({
        'status': 'success',
        'failed_nodes': failed_count,
        'stats': stats,
        'recovery_log': network.recovery_log[-5:] # Last 5 log entries
    })

@app.route('/find_path', methods=['POST'])
def find_path():
    if not network:
        return jsonify({'error': 'Network not initialized'})

    data = request.json
    start = data.get('start')
    end = data.get('end')

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if start is None or end is None:
    return jsonify({'error': 'Start and end nodes required'})

start_time = time.time()
primary_path = network.dfs_find_path(int(start), int(end))
alternate_paths = network.find_alternate_paths(int(start), int(end))
path_time = time.time() - start_time

return jsonify({
    'primary_path': primary_path,
    'alternate_paths': alternate_paths,
    'computation_time': path_time,
    'path_found': primary_path is not None
})

@app.route('/recover_network', methods=['POST'])
def recover_network():
    if not network:
        return jsonify({'error': 'Network not initialized'})

    recovery_time = network.recover_network()
    stats = network.get_network_stats()

    return jsonify({
        'status': 'success',
        'recovery_time': recovery_time,
        'stats': stats,
        'recovery_log': network.recovery_log[-10:] # Last 10 log entries
    })

@app.route('/remove_duplicates', methods=['POST'])
def remove_duplicates():
    if not network:
        return jsonify({'error': 'Network not initialized'})

    original_edges = len(network.edges)
    network.remove_duplicates()
    removed = original_edges - len(network.edges)

    stats = network.get_network_stats()

    return jsonify({
        'status': 'success',

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        'duplicates_removed': removed,
        'stats': stats,
        'edges': network.edges
    })

@app.route('/reset_network', methods=['POST'])
def reset_network():
    global network
    if network:
        network.failed_nodes.clear()
        network.failed_edges.clear()
        network.recovery_log.clear()

        # Rebuild DSU
        network.dsu = DisjointSetUnion(network.num_nodes)
        for u, v, w in network.edges:
            network.dsu.union(u, v)

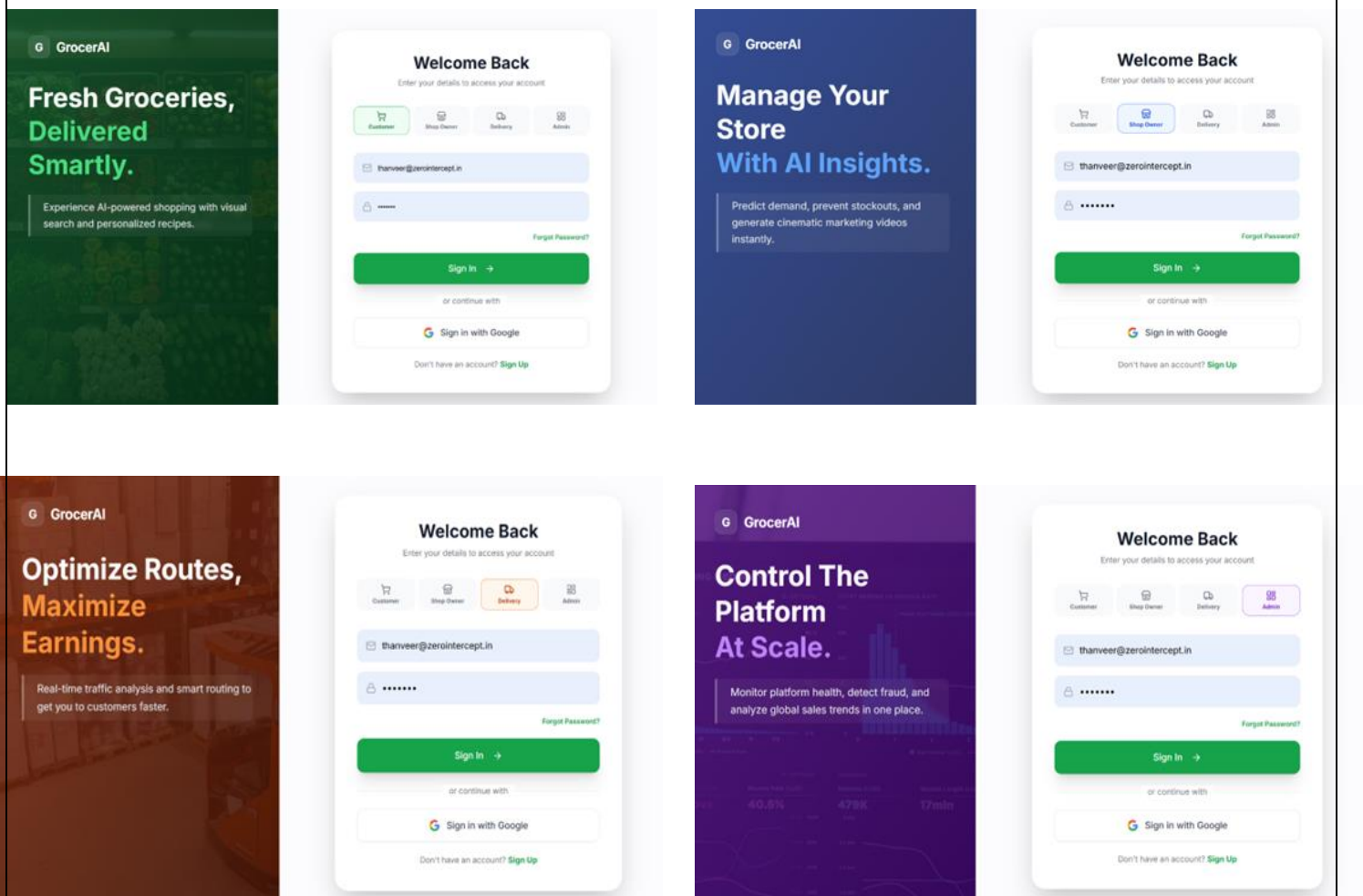
    stats = network.get_network_stats() if network else {}
    return jsonify({'status': 'success', 'stats': stats})

@app.route('/get_stats', methods=['GET'])
def get_stats():
    if not network:
        return jsonify({'error': 'Network not initialized'})

    stats = network.get_network_stats()
    return jsonify({
        'stats': stats,
        'failed_nodes_list': list(network.failed_nodes),
        'recovery_log': network.recovery_log,
        'edges': network.edges
    })

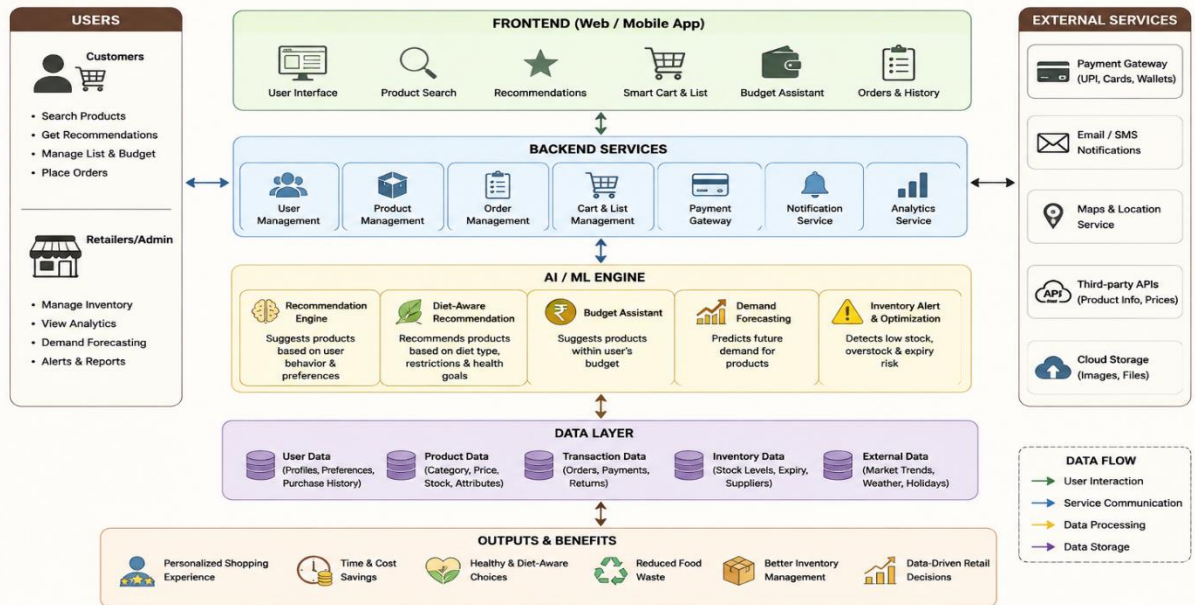
```


Output:



AI-Powered Smart Grocery Shopping System

System Architecture



AI-POWERED SMART GROCERY SHOPPING SYSTEM

